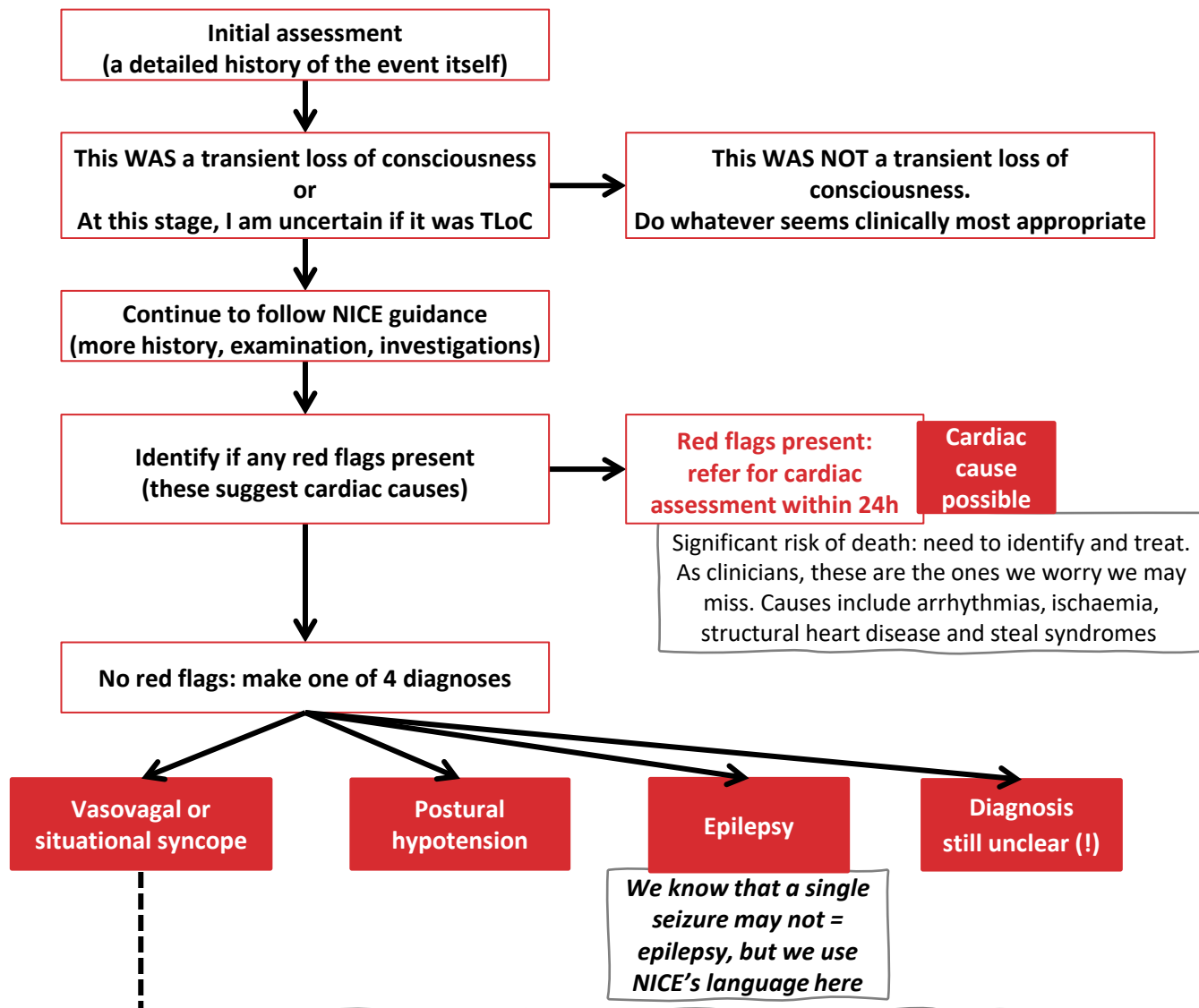




NICE transient loss of consciousness (TLoC) guidance encourages us to take the following approach (if $\geq 16y$).

(The details of each step are given in sections 2 and 3 of this GEMS.)



An old but useful BMJ review reminds us (BMJ 2010:341:c4457):

- Vasovagal or situational syncope is sometimes referred to as neurally-mediated: characterised by vasodilatation or vagal stimulation, causing bradycardias or (brief, recoverable) asystole, causing collapse.
- Remember that the older you are, the less likely the cause is vasovagal/situational syncope (and benign).** 90% of syncope in those $<40y$ is vasovagal/situational, but only about 50% is in those $>60y$.
- Although the cause is benign, the event can result in harm (e.g. fractured hip from fall).**

Safety advice for all with transient loss of consciousness

DRIVING



Those with transient loss of consciousness must follow DVLA guidance (section 4 of this GEMS). Those waiting for specialist assessment must not drive until told they are safe to do so.

HEALTH AND SAFETY AT WORK: note patient's occupation. Discuss the implications of TLoC on work, and any actions they must take to ensure the safety of themselves and others (as per Health and Safety at Work Act).



Initial assessment: history, examination, investigations

Take a detailed history from patient and witnesses of the circumstances of event, including:

Beforehand	During the event	Afterwards
- Posture immediately before - Prodromal symptoms (sweating, feeling warm/hot)	- Appearance (e.g. eyes open/shut, colour) - Presence or absence of movement during event (e.g. limb jerking and duration) - Tongue biting - Injury during event	- Presence or absence of confusion during recovery - Unilateral weakness during recovery - Duration (from onset to regaining consciousness)

A note on tongue biting: NICE says record if lateral or tip biting: lateral tongue biting may be a better predictor of epilepsy than tip biting, but the evidence for this comes from very highly-selected populations so record all biting.

At this stage, you may be able to rule out TLoC, identify a diagnosis and manage appropriately.

If you think TLoC has occurred, OR you remain unsure if this was TLoC, proceed to the next step.

Take a detailed history of:

- Previous similar events (description, number, frequency).
- PMH/family history of cardiac events (especially sudden cardiac death).
- Current medications, including any that may contribute to TLoC (e.g. diuretics).

Examination

- Record vital signs (P, BP, RR, T).
- Lying and standing BP if clinically indicated (see below for how to do this).
- Cardiovascular and neurological examination.
- Any relevant examination triggered by history.

Investigations

- ECG, preferably with automated interpretation.** Treat as red flag if any ST segment or T-wave changes, long or short QT interval or conduction abnormalities (e.g. complete R or LBBB, any degree of heart block).
- Do any other relevant tests** (e.g. blood sugar measurement in diabetics, Hb if anaemia/bleeding suspected).
- Do NOT routinely request an EEG.

ECGs: NICE advises automated interpretation. If using an **unautomated ECG**, the clinician must be confident identifying: persistent bradycardia, ventricular ectopics and arrhythmias, corrected QT interval, Brugada syndrome, ventricular pre-excitation, ventricular hypertrophy, T-wave inversion, pathological Q-waves, sustained atrial arrhythmias, paced rhythms.

Measuring lying and standing BP (as per NICE hypertension guidance, NG136, 2023)

- Check BP lying down (NICE doesn't say for how long to lie for) **or** seated if inconvenient to measure BP lying.
- Measure BP again after the person has been standing for at least 1 minute.
- **A drop of either 20mmHg in SBP or 10mmHg in DBP is significant.** If BP drop NOT significant BUT symptoms suggestive of postural hypotension and BP was done seated, repeat after lying down.
- **If postural hypotension diagnosed: follow section 3 of GEMS (on postural hypotension).**
- **If BP drop NOT significant BUT symptoms suggestive of postural hypotension: refer cardiology.**

Identify if any red flags present (these suggest cardiac cause):

Cardiac cause possible

From history/examination:

- TLoC during exertion.
- ECG abnormality (as described above).
- Heart murmur.
- Heart failure (history or signs/symptoms).
- New or unexplained breathlessness.

Past medical/family history:

- Family history of sudden cardiac death <40y.
- Known inherited cardiac condition.

NOTE: syncope during exercise makes cardiac cause likely. Syncope shortly after stopping makes vasovagal cause more likely.

- **If red flags: cardiology assessment within 24h.**
- **Consider assessment within 24h if ≥ 65 y with TLoC without prodromal symptoms.**
- **Arrange appropriate urgent assessment if indicated for another reason (e.g. fracture during TLoC).**

Once red flags eliminated, 1 of 4 possible diagnoses should be made:

Vasovagal/situational syncope

Postural hypotension

Epilepsy*

Diagnosis still unclear

* We know that a single seizure may not = epilepsy but we use NICE's language here

NICE on transient loss of consciousness

3. Diagnostic options

NICE 2023, CG109



Red Whale

GEMS

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Based on history, examination and investigations, make 1 of 4 possible diagnoses:

VASOVAGAL or SITUATIONAL SYNCOPE	VASOVAGAL syncope (faint) likely if No features of alternative diagnosis <u>AND</u> the 3Ps may be present: P rovoking factors e.g. pain, medical procedure P osture e.g. prolonged standing or previous episodes aborted by lying down P rodromal symptoms e.g. sweating, feeling hot before event SITUATIONAL SYNCOPE diagnosed if No features of alternative diagnosis <u>AND</u> TLoC consistently provoked by doing the same thing. Common triggers are: straining during micturition (usually when standing), laughing, coughing, Valsalva-induced (weightlifter's/trumpet blower's syncope) (BMJ 2010:340:c880). If VASOVAGAL or SITUATIONAL SYNCOPE likely: - Reassure patients and explain mechanism of syncope. - Advise people to keep record of symptoms/events to help them understand/avoid triggers. - Re-consult GP if recurs, especially if any features are different from usual events. - DVLA guidance – see section 4 of this GEMS. <i>(If presentation is not to the GP, the patient should be advised to take a copy of the clinical report and ECG to the GP. If an ECG has not been done, the GP should arrange this within 3d.)</i>
POSTURAL HYPOTENSION	POSTURAL HYPOTENSION suspected if No features suggesting alternative diagnosis <u>AND</u> history is typical e.g. postural dizziness/falls Assess for postural BP drop (see previous page for how to do this). A drop of either 20mmHg in SBP or 10mmHg in DBP is significant. Remember: if symptomatic but drop not significant when done seated, do a lying/standing BP. ➤ If drop NOT significant BUT symptoms suggestive of postural hypotension: refer cardiology. If POSTURAL HYPOTENSION DIAGNOSED - Consider causes, including medication. - Manage appropriately (e.g. advice on preventing falls). - Measure subsequent BPs standing. - Consider specialist referral if symptoms persist despite addressing likely causes.
EPILEPSY	EPILEPSY suspected if 1 or more features that are strongly suggestive of seizure, which are: - Prodromal déjà vu or jamais vu. - Witness reports abnormal behaviour during event <i>which patient has no memory of.</i> <u>Prolonged</u> limb jerking (vasovagal syncope can be associated with brief limb jerking). - Tongue biting. - Head turned to one side during TLoC. - Unusual posturing. - Confusion after event. If EPILEPSY SUSPECTED Refer to specialist for assessment within 2w. Advise not to drive (see DVLA, section 4). Epilepsy less likely if any of the following occur: - Prodromal symptoms that, in the past, have been aborted by sitting/lying down. - Sweating before an episode. - Precipitated by prolonged standing. - Pallor during episode.
DIAGNOSIS STILL UNCLEAR	Refer ALL with TLoC for specialist cardiovascular assessment UNLESS uncomplicated vasovagal/situational syncope, postural hypotension OR being referred on epilepsy pathway.

We make every effort to ensure the information in these pages is accurate and correct at the date of publication, but it is of necessity of a brief and general nature. The information presented herein should not replace your own good clinical judgement, or be regarded as a substitute for taking professional advice in appropriate circumstances. In particular, we suggest you carefully consider the specific facts, circumstances and medical history of any patient, and recommendations of the relevant regulatory authorities. We also suggest that you check drug doses, potential side-effects and interactions with the British National Formulary. Save insofar as any such liability cannot be excluded at law, we do not accept any liability for loss of any type caused by reliance on the information in these pages. March 2024. For full references see the relevant Red Whale articles.

Transient loss of consciousness

4. DVLA guidance

Assessing fitness to drive – a guide for medical professionals, Dec 2023

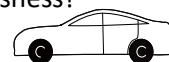


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GEMS
Guidelines & Evidence Made Simple

DVLA guidance is COMPLICATED and updated from time to time. Use below as a guide but ALWAYS confirm with 'DVLA: assessing fitness to drive – a guide for medical professionals' online before advising a patient.

- For syncope, decide which type of syncope you are dealing with as the DVLA has multiple categories: these are explained in the relevant row headings in the table below!
- Note that for RECURRENT typical vasovagal syncope, the DVLA asks us to estimate the likelihood of the annual risk of recurrence. I have no idea how we do this!
- NICE talks about THREE 3Ps to identify vasovagal syncope: the DVLA highlights that 2 of these are particularly important when it comes to driving:
 - PRODROME: is there sufficient warning to find a safe place to stop before losing consciousness?
 - POSTURE: episodes that only occur when standing are less of a concern to the DVLA.



	Group1 (car and motorcycle)	Group 2 (bus and lorry)
Cough syncope (multiple episodes over 24h count as a single event, but episodes >24h apart count as separate episodes)		
Any posture	- Must NOT drive, must notify DVLA. - Must not drive for: 6m following a single episode. 12m following multiple episodes over 5y.	- Must NOT drive, must notify DVLA. - Must not drive for: 12m following a single episode. 5y following multiple episodes over 5y.
Typical vasovagal syncope (single episode or recurrent episodes with identifiable consistent prodrome)		
While standing	May drive, no need to notify DVLA.	Must NOT drive, must notify DVLA.
While sitting	- May drive. - No need to notify DVLA PROVIDED avoidable trigger which will not occur when driving: otherwise, must not drive until annual risk of recurrence is assessed to be <20%.	- Must NOT drive, must notify DVLA. - Investigate for identifiable/treatable cause. - If single episode: must not drive for 3m. - If recurrent with identifiable consistent prodrome, in addition to the above: must not drive until annual risk of recurrence is below 2%.
Syncope with avoidable trigger or reversible cause (single or recurrent episode)		
While standing	May drive, no need to notify DVLA.	Must NOT drive, must notify DVLA.
While sitting	- Must not drive for 4w. - Driving may resume after 4w ONLY if cause identified and treated. - Must notify DVLA if cause has not been identified/treated.	- Must NOT drive for 3m, must notify DVLA. - Driving may resume after 3m ONLY if cause has been identified and treated. - Must notify DVLA if cause has not been identified/treated.
Unexplained syncope, including syncope without reliable prodrome (DVLA says this diagnosis applies only after appropriate neurological/cardiac investigations have found no abnormality)		
Any posture	- Must NOT drive, must notify DVLA. - If no cause found, licence will be refused/revoked for 6m after single episode, 12m if recurrent.	- Must NOT drive, must notify DVLA. - If no cause found, licence will be refused/revoked for 12m for single episode, 10y if recurrent.
Cardiovascular, excluding typical syncope		
Any posture	- Must not drive, must notify DVLA. - Driving may resume after 4w ONLY if cause identified and treated. - If no cause found, licence will be refused/revoked for 6m if single episode, 12m if recurrent.	- Must NOT drive, must notify DVLA. - Driving may resume after 3m if cause has been identified and treated. - If no cause found, licence will be refused/revoked for 12m if single episode, 2y if recurrent.
Blackout with seizure markers: this is for those where, on the balance of probability, there is clinical suspicion of a seizure but no definite evidence. Must have specialist assessment/investigation.		
Any posture	- Must NOT drive, must notify DVLA. - Must not drive for 6m. May be increased to 12m if factors that may increase risk of recurrence. If recurrent, follow DVLA epilepsy guidance.	- Must NOT drive, must notify DVLA. - Must not drive for 5y. - If recurrent, follow DVLA epilepsy guidance.

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